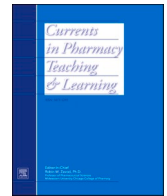




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Research Paper

Using an educator-guided generative artificial intelligence (GenAI) tool for developing communication skills in undergraduate pharmacy students



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ABSTRACT

Introduction: Effective communication of health messages is essential. Pharmacy educators face the challenge of adapting curricula or integrating Generative AI (GenAI) tools. This study presents an educator-guided GenAI prototype, Authentic Teaching & Learning Applications Simulation (ATLAS), which offers real-time simulations with patient avatars and provides both verbal and non-verbal feedback to students.

Methods: A quasi-experimental study was conducted with pharmacy undergraduates. One month before their summative Objective Structured Clinical Examination (OSCE), students accessed ATLAS, which simulates patient interactions and delivers immediate feedback. Impact was evaluated through usage data, OSCE scores and student perceptions. The primary outcome was the difference in change in OSCE communication scores from previous OSCE between ATLAS users (201) and non-users (313; a control group).

Results: For students who had two data points for the primary outcome analysis, ATLAS users (188) showed significant improvement in their communication scores (median increase = 1.55 points, $p = 0.0045$), while controls (273) showed no change between years ($p = 0.7$). When comparing conversation attempts to OSCE scores in the ATLAS group ($n = 201$), improvements were seen with three or more attempts: three (+6.7), four (+6.5), five (+8.3), and six (+7.7). Domestic and female students benefited most. Students reported that GenAI avatars were realistic and enhanced learning, offering authentic, safe, and accessible practice with personalised feedback.

Discussion: Despite some technical limitations and AI hesitancy, ATLAS was a valuable supplement. The study supports that different communication subskills may be differentially associated with technology supported practice, and students who perhaps are not as confident to participate in the classroom could benefit from GenAI practice prior to classroom practice. However, many students still preferred in-person practice for developing empathy and nuanced communication.

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Conclusion: The present study provides associational evidence that educator guided practice with ATLAS correlates with higher OSCE communication scores, particularly for repeated users, amid notable student engagement and perceived value. Future research should prioritise multi-site randomised trials to minimise self-selection bias with explicit plans to assess equity impacts (e.g., for international students).

Introduction

Academics have a role in ensuring pharmacy graduates have competent clinical communication prior to entering the workforce.^{1,2} Simulating real world settings so students can genuinely practice their communication skills sometimes involves high-cost formative objective structured clinical examination (OSCEs)³ or more recently can involve using virtual simulated patients (VSPs).⁴ With the introduction of Generative Artificial Intelligence (GenAI) education tools, there is now the opportunity to provide a cost effective, customised learning experience, showing potential to develop skills for patient counselling within pharmacy education.^{5–7}

GenAI has raised concerns in healthcare disciplines about diminishing students' need to think critically and possibly weakening their communication abilities,⁸ suggesting well-designed AI tools, steered by teachers with pedagogical skill and intent, are needed to optimise the learning use of GenAI.^{9,10} There is growing recognition of the need for frameworks that systematically integrate AI capabilities to support authentic learning experiences while limiting various forms of bias and 'hallucinations'.¹¹ Rather than relying solely on large language models (LLMs), effective educational systems require carefully designed architectures built on theory and evidence. Before the rise of LLMs, developing virtual simulated patients required advanced coding skills and capabilities due to the complexity of creating such platforms.¹¹ With the launch of ChatGPT, efforts have been made to leverage LLMs to create virtual patients without the need for complex coding striving to improve authenticity, adaptability, and transferability across healthcare domains. Although some chatbots for training OSCEs already exist,^{12–16} there is limited evaluation data on the impact of these tools on learning outcomes. Current research landscape primarily examines student perceptions and tool feasibility.^{9,12–16}

Numerous platforms utilise LLMs to create VSPs. Geeky Medic, OSCER AI, and Curefun solely provide text and verbal interactions without incorporating non-verbal cues that are integral to authentic clinical communication. This study aims to investigate the impact of a GenAI simulation prototype, Authentic Teaching & Learning Applications Simulation (ATLAS) on clinical communication in undergraduate students. ATLAS simulates patient interactions within a pharmacy setting, providing students with opportunities to practice clinical communication and decision-making in a realistic and safe environment, providing both verbal and non-verbal feedback, allowing students to have an authentic learning experience, similar to a formative OSCE.

Materials and methods

Study design

We conducted a quasi-experimental study comprising of a non-equivalent group design comparing students who used ATLAS and who did not use ATLAS. Both quantitative and qualitative data were collected using back-end user ATLAS data, OSCE grades and a

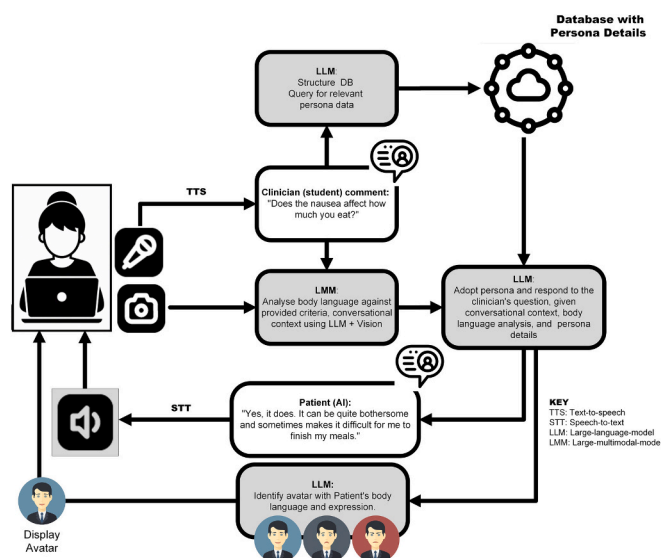


Fig. 1. Large Language Model workflow of ATLAS (Authentic Teaching & Learning Applications Simulation).

cross-sectional survey with open and closed ended questions investigating perceptions of use. The following analyses were included: 1) a longitudinal analysis investigating the change in OSCE communication score from the previous year (defined as baseline OSCE score) between ATLAS users and non-ATLAS users; 2) a cross sectional analysis of OSCE grades and ATLAS attempts between ATLAS users and ATLAS non-users (zero attempts); 3) an analysis of both open and closed ended survey questions in the ATLAS group only.

Participants

The study was conducted at a major tertiary institution in Australia. In 2024, all second-year students had access to ATLAS. Students who chose not to use ATLAS as well as those who completed the same OSCE the year before (without ATLAS) served as controls. For the longitudinal analysis, OSCE communication scores from the previous year were retrieved for both groups. Students who did not have two sets of OSCE scores for comparison were excluded.

GenAI prototype, Authentic Teaching & Learning Applications Simulation (ATLAS)

The GenAI prototype, ATLAS, utilises frontier large language models from Anthropic (Claude Sonnet) and OpenAI (GPT-4) as its conversational and review engines. The system employs a multi-model architecture: Claude Sonnet serves as the primary conversation engine for avatar interactions and feedback generation, while GPT-4 Vision provides body language analysis. ATLAS integrates speech recognition, natural language processing, text-to-speech, and body language analysis to deliver an authentic simulation experience (Fig. 1). The conversational AI operates through structured prompts defining role specifications, communication parameters, and emotional response constraints based on educator-defined criteria. AI-generated (Midjourney V6.1) avatars represent multiple emotional states, with dynamic selection based on conversation context. Feedback generation aggregates conversation transcripts and body language data, analyses them against educator-defined evaluation criteria, and generates structured feedback addressing performance, strengths, and areas for improvement. Data handling employs a secure graph database with session-based authentication; ephemeral visual data is processed in real-time and deleted immediately, while conversation transcripts are retained for student reflection. Educator oversight is maintained through multiple mechanisms: educators define avatar design parameters, evaluation criteria aligned with learning objectives, and conversation structure. Avatars are tested extensively by educators. Ethical safeguards restrict feedback algorithms to educator-specified criteria with multiple validation layers preventing criterion drift (see Appendix 1 for detailed technical specifications). ATLAS has been successfully piloted in nutrition and dietetics education using GenAI patients to bridge theory and practice by supporting real world communication skills training for students.¹⁷ A video of our design for pharmacy is available at <https://www.youtube.com/watch?v=siWNJfyF4ZY>.

OSCE design

The OSCEs are conducted annually in the Bachelor of Pharmacy program. While students are assessed on different clinical competencies across stations each year, the same communication checklist (Appendix 2) is used throughout the degree to ensure continuity and consistency in the development of communication skills. In the second-year summative OSCE, students complete two stations: a 15-min history-taking station and an 8-min primary care station, both involving simulated patients in a community pharmacy setting. In the history-taking station, students gather medication-related information to support reconciliation and produce an accurate written medication list. Each OSCE case had a combination of potential errors (e.g., patient no longer taking some medication in the bag they brought in, or patient was taking medication that was not brought in with them to the pharmacy). This station includes verbal and written components, with its design published elsewhere.¹⁸ In the primary care station, simulated patients present requesting a particular product and students have to determine if the product is safe to supply based on the clinical information they elicit from the patient. Assessment in both stations includes communication skills (e.g., fluency, empathy, structure) and clinical competence (e.g., accuracy of medication list), weighted equally. Examiners and cases were based on a roster and managed per session. Examiner variability and case difficulty were managed using an Item Response Model, and all examiners received routine training and calibration on the communication rubric, with standardized checklists used across stations. Examiners were unaware of students' ATLAS usage status. While cases are different, they have the same checklist structure with same learning expectations, and only vary in clinical content area. Overall, OSCE procedures, scoring criteria, and examiner preparation were consistent across cohorts, with variation in cases and examiners reflecting normal curriculum delivery.

ATLAS implementation and case design

In 2024, ATLAS was released to students to access one month prior to the OSCEs. Engagement with ATLAS was optional with no incentives. ATLAS practice cases were drawn from existing OSCE cases and piloted by academics and postgraduates to ensure avatar responsiveness. Prior to the introduction of ATLAS, students practised OSCE scenarios through peer role plays in facilitated workshops. These workshops have continued post-ATLAS, providing opportunities for safe practice and skill reinforcement.

Outcomes

Our primary outcome was the change in OSCE communication scores between the ATLAS group and control group (non-ATLAS users) from their previous year's OSCE score. Our secondary outcomes were improvement in OSCE communication scores with each

ATLAS attempt, student engagement perception data and the factors that affected engagement with ATLAS.

Data collection

OSCE grades from communication checklists were collected from the assessment databases. User attempts were taken from ATLAS. All students who engaged with ATLAS were asked to complete a 15-min perception survey on their learning management system, which included fixed Likert scale responses. In addition, students were asked to do a 10-min open-ended written reflection exploring how they used ATLAS and their reasons for engaging with it.

Data analysis

Descriptive and exploratory analyses were performed using GraphPad Prism (V. 10.3.1). Data were not normally distributed, hence non-parametric tests were used. The relationship between ATLAS usage and communication scores was assessed using Kruskal-Wallis test with post-hoc Dunn's test, comparing scores of students with 1–6 conversations to those with no conversations. Dunn's method adjusts *p*-values for multiple comparisons (family-wise $\alpha = 0.05$). For groups showing significant differences in the Kruskal-Wallis test, pairwise Mann-Whitney *U* tests were conducted with Dunn's correction for multiple comparisons, and rank-biserial correlation (*r*) was calculated as a non-parametric effect size measure. To analyse the relationship between dialogue count (total number of student-avatar conversational exchanges, where each exchange consists of one student turn and one avatar response) and communication scores, linear regression with 95% confidence intervals was performed to assess whether slopes significantly deviated from zero. Results are presented as means with standard error of the mean (SEM) where appropriate.

Beta regression models were implemented in RStudio¹⁹ using the betareg package²⁰ to analyse OSCE performance data. This approach is ideal for percentage-based outcomes (0–100%) because it respects natural boundaries of the data and provides accurate parameter estimates. Models were built using a stepwise approach with selection based on corrected Akaike Information Criterion (AICc). The models included ATLAS usage (users vs. non-users), gender (male vs. female, reference: female), student status (domestic vs. international, reference: international), baseline OSCE performance from OSCE in the previous year (final_previous and comm_previous scores), and interaction terms (gender $\times$ ATLAS use, student status $\times$ ATLAS use, student status $\times$ baseline, gender $\times$ student status). Multiple candidate models were tested for each outcome, with the best model selected by lowest AICc. Model fit was assessed using pseudo R^2 and residual diagnostics. Results are reported as beta coefficients (β) with 95% confidence intervals and *p*-values. Detailed model parameters and fit statistics are provided in Appendix 3 (Supplementary Tables S1-S2).

Open-ended questions from student reflections were analysed deductively using latent content analysis²¹ and grouped under Fredricks' three-dimensional engagement model²² with AI-assisted theme generation. Fredricks' model proposes that students' engagement is presumed to be malleable, responsive to contextual features, and amenable to environmental change. Engagement was conceptualised as multifaceted where behavioural engagement reflects the effort students believe it is necessary for academic success; emotional engagement encompasses positive and negative interactions that influence willingness to do the work; and cognitive engagement considers the effort required to master different skills. Using OpenAI's GPT-4o model via API, the analysis followed a two-stage process (see Appendix 4 for complete prompts and code): (1) the AI generated five independent sets of 4–5 potential themes aligned with Fredricks' three dimensions from the complete dataset of student reflections (temperature = 1.8, $n = 5$ completions), and (2) the AI evaluated and selected the best five non-overlapping themes mapped to behavioural, emotional, and cognitive engagement. Student reflections were anonymised prior to AI analysis with only de-identified student numbers retained for linking data. OpenAI API data retention policies specify that data submitted via API is not used for model training. Two researchers (AM, AL) independently reviewed the AI-generated themes.¹⁷ Reflexivity was applied through a reader-response exercise, in which AM and AL reviewed examples and negotiated any biases regarding student perspectives. Incomplete surveys were considered for any questions they answered, but responses left blank were excluded. Researcher (AL) then manually coded all 201 individual student responses with consultation with the team, using the validated themes as a framework within Fredricks' three-dimensional model. When coding, researchers interpreted the learning environment as the students' own personal space outside of the classroom, and as a safe space, non-judgmental setting where mistakes could be made without worry. This risk-free environment to practice clinical communication and being able to do so in their own environment, without the need for peer or facilitator input, helps develop professional growth. A cultural insight used was to represent a shift in pharmacy education from reliance on peer/facilitator input alone for simulating clinical conversations to now being able to utilise digital simulation in an effective manner. The ATLAS platform helps mediate professional identity formation whereby students can begin to take on the persona of a pharmacist in an AI-powered environment. Researchers aim to investigate how practice with ATLAS was embedded in the symbolic and cultural world of pharmacy training, not just isolated behaviors.

Ethics approval

This project was approved by the institution's Human Research and Ethics Committee (Project no: 41702).

Results

Demographics of students

A total of 514 second-year students who completed the same OSCE were included in the analysis which included 201 who used the platform at least once, and 313 students who never used ATLAS who served as controls. Table 1 shows demographics of students.

Longitudinal OSCE performance analysis

For the longitudinal analysis, we back-tracked both ATLAS and non-ATLAS users to obtain their OSCE score from the previous year. Of these, there were 188 students from ATLAS users and 273 students from non-ATLAS users included in the longitudinal analysis. Fig. 2 shows a flowchart of participants. Fig. 3 shows the median change in OSCE scores from the previous year's OSCE between ATLAS and non-ATLAS users. ATLAS users showed significant improvement in their communication scores (median increase = 1.55 points, $p = 0.0045$), while controls showed no change between years ($p = 0.7$).

ATLAS usage frequency and OSCE performance outcomes

Among the ATLAS users, engagement showed an engagement-performance relationship with OSCE communication scores (Fig. 4 and Appendix 3 Supplementary Table S3). Usage patterns among the 201 users were: one conversation ($n = 35$), two conversations ($n = 23$), three conversations ($n = 48$), 4 conversations ($n = 31$), five conversations ($n = 27$), and six or more conversations ($n = 37$; See Fig. 5.)

For students who completed 6 or more conversations, the average number of interactions was 8 (SD = 5.27). Compared to non-users who had a mean OSCE communication score of 86.0, improvements were observed for students who completed three or more conversations (Fig. 4). Kruskal-Wallis test showed significant differences across groups ($H = 37.63$, $df = 6$, $p < 0.0001$, $n = 514$). Results were as follows: one conversation ($n = 35$, mean = 90.4, +4.4 points, $Z = 1.30$, adjusted $p > 0.99$); two conversations ($n = 23$, mean = 91.8, +5.8 points, $Z = 1.99$, adjusted $p = 0.28$); three conversations ($n = 48$, mean = 92.6, +6.6 points, $Z = 3.72$, adjusted $p = 0.0012$); four conversations ($n = 31$, mean = 92.4, +6.4 points, $Z = 2.66$, adjusted $p = 0.047$); five conversations ($n = 27$, mean = 94.3, +8.3 points, $Z = 3.61$, adjusted $p = 0.0019$), and six or more conversations ($n = 37$, mean = 93.8, +7.8 points, $Z = 3.63$, adjusted $p = 0.0017$). Effect size analyses (rank-biserial correlation) for significant comparisons showed $r = 0.32$ (3 conversations), $r = 0.29$ (4 conversations), $r = 0.41$ (5 conversations), and $r = 0.36$ (6+ conversations), indicating small to medium effects.

To better understand the nuanced impact of ATLAS on different assessment components and student demographics, Beta regression models were fitted controlling for baseline OSCE performance (from previous year), gender, and student status (complete model parameters in Appendix 3 Supplementary Tables S1-S2). After adjusting for these covariates, ATLAS use was significantly associated with improved primary care communication scores ($\beta = 0.37$, 95% CI [0.13, 0.61], $p = 0.003$). However, the main effect of ATLAS on history-taking communication was not significant after controlling for baseline differences ($\beta = -1.10$, $p = 0.334$). Baseline OSCE performance was a significant predictor across all outcomes ($p < 0.001$), and domestic students consistently demonstrated higher scores than international students (history-taking: $\beta = 1.84$, $p = 0.039$; primary care: $\beta = 2.19$, $p = 0.025$; overall: $\beta = 2.65$, $p = 0.002$). Importantly, significant gender $\times$ ATLAS interactions revealed that male students benefited less from ATLAS than female students in primary care communication ($\beta = -0.49$, $p = 0.017$), with a similar marginally significant trend in history-taking communication ($\beta = -0.39$, $p = 0.073$). These improvements hold educational significance, as each additional point on the rubric reflects the use of more verbal strategies to enhance interaction between interlocutors.

Student quantitative perception data

Survey results indicated students found ATLAS useful (Fig. 6): 80.0% ($n = 160$) agreed or strongly agreed that they would like more scenarios in this interactive style, and 67.5% ($n = 135$) found the simulation engaging. The majority (80.4%; $n = 160$) agreed that ATLAS allowed them to practise history taking in an interactive manner, and 63.5% ($n = 127$) agreed that it helped develop patient-friendly language skills. Additionally, 68.3% ($n = 136$) reported that ATLAS provided meaningful feedback on their communication

Table 1
Student demographics.

Characteristic	ATLAS Users ($n = 201$)	Controls ($n = 313$)
Gender, n (%)		
Female	138 (68.7)	208 (66.5)
Male	61 (30.3)	105 (33.5)
Did not disclose	2 (1.0)	0 (0.0)
Student Status, n (%)		
Domestic	130 (64.7)	183 (58.5)
International	71 (35.3)	130 (41.5)

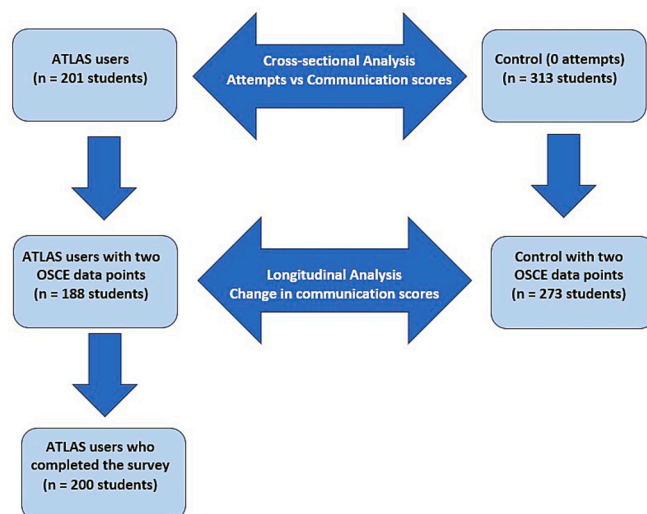


Fig. 2. Participant flow chart.

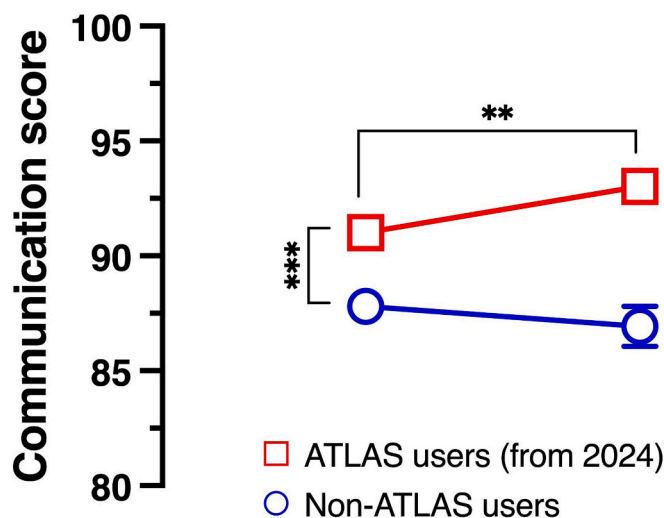


Fig. 3. Longitudinal comparison of OSCE communication scores. ATLAS users ($n = 188$; squares) and non-users ($n = 273$; circles) showing change in OSCE scores. Data shown as mean $\pm$ 95% confidence intervals. Wilcoxon signed-rank test used for paired comparisons within groups and Mann-Whitney test used for unpaired comparison between groups.

skills, and 59.0% ($n = 118$) reported that it helped maintain their interest in the activity. Students also indicated that ATLAS improved their approach to empathetic patient interactions (47% agreed or strongly agreed; $n = 94$). These results were based on a 5-point Likert scale survey completed by 200 of the 201 ATLAS users, though response rates varied slightly by question ($n = 199$ –200).

While the majority of responses were positive, some aspects of the ATLAS simulation received more neutral or less enthusiastic feedback. Only 40.5% of students agreed that computer-generated criticisms were a less offensive way to receive feedback compared to facilitator feedback. Additionally, 50.5% agreed that the simulation was realistic, indicating room for improvement in perceived authenticity, though this still represents a majority of users finding the experience realistic.

Student qualitative perception data

Overall, the students' engagement in the tool was high and can be mainly attributed to the convenience and efficiency of using ATLAS for repeated practice for a high stakes assessment without the need for asking peers or family members to support their practice. There was also a sense of students not wanting to miss out, especially if all their peers were using it for practice. Table 2 highlights some illustrative quotes mapped to the three-dimensional student engagement model.²²

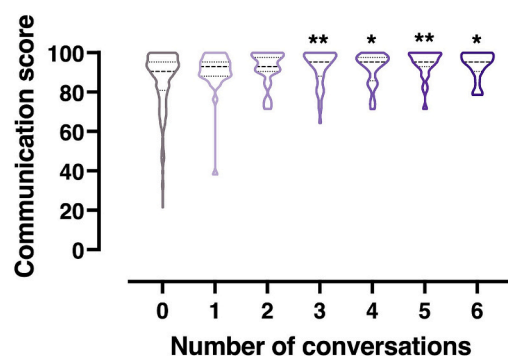


Fig. 4. Communication scores by number of ATLAS conversations. Controls ($n = 313$) serving as zero attempts shown in grey, while students who completed 1–6+ conversations are shown in purple (n ranges from 23 to 48 per group). Kruskal-Wallis test with post-hoc Dunn's test compared each usage group to controls ($*p < 0.05$, $**p < 0.01$). Violin plots show score distributions with medians (dashed lines) and quartiles (dotted lines). Means and 95% confidence intervals for each group are provided in Supplementary Table S3. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

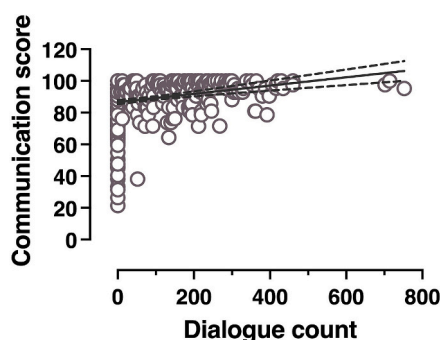


Fig. 5. Communication scores as a function of dialogue count. Purple circles represent individual student data. While the data shows non-parametric distribution, a simple linear regression fit is shown (solid black line) with 95% confidence intervals (dashed lines) to illustrate the general positive trend (slope = 0.026, $R^2 = 0.059$, $p < 0.0001$), though the relationship may not be strictly linear. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Behavioural engagement

As ATLAS was preparing students for a high stakes assessment, students embraced engaging with the tool in order to achieve academic success, mainly attributing success to more attempts and more practice. Students wanted to utilise all possible methods to practise for the OSCE; “ATLAS was a must use tool for extra OSCE practice.” Student 203. Indeed, ATLAS was seen as an additional tool for practice and not a replacement, with students citing that to maximise OSCE preparation, they needed to use a combination of peer-to-peer practice and ATLAS use; “every aspect of practice was effective especially doing the cases given to us by the university and, ATLAS and workshop scenarios with friends” Student 70. The repeated practice built up the confidence of students for the actual OSCE; “I felt more comfortable going into the actual OSCE” Student 211. Students who seemed to benefit the most from ATLAS use were students who stated they were less extroverted, and appreciated the privacy of practising online in their own time to improve their skills before practising in front of others; “because I could do it in the comfort of my own home without the pressure of classmates watching” Student 214.

Emotional engagement

Many students felt awkward or uncomfortable interacting with an AI initially. Nonetheless, repeated use helped some students feel more comfortable and confident, highlighting a learning curve in adapting to the AI; “At first, it was strange that the patient's facial expressions changed depending on what question I asked, but once I got use to this, I felt this aspect was good to have” Student 236. ATLAS was seen to be able to bring real life interactions to the screen, “Pretty amazing, as it can react to any of my questions” Students 103. While ATLAS provided a valuable supplementary practice tool, many students felt that in-person practice with peers or facilitators was more effective for developing empathy and nuanced communication skills; “no computer can replicate a real patient” Student 230, and that ATLAS avatars often provided lengthy responses which overwhelmed students; “It gave a lot more information than you would normally get from a patient” Student 94; “It exceeded my expectations, though there are still many improvements needed for the AI software to be a better representation of the real interaction” Student 99. Technical hiccups deterred students from finishing off their attempts as well as the fixed patient cases, with students noting they can make a wider range of cases with their friends, “I practiced a LOT with friends, but I also used ATLAS. My friends and I came up with our own patient scenarios with different personalities for each so it was as unique as possible” Student

How much do you agree with the statement?

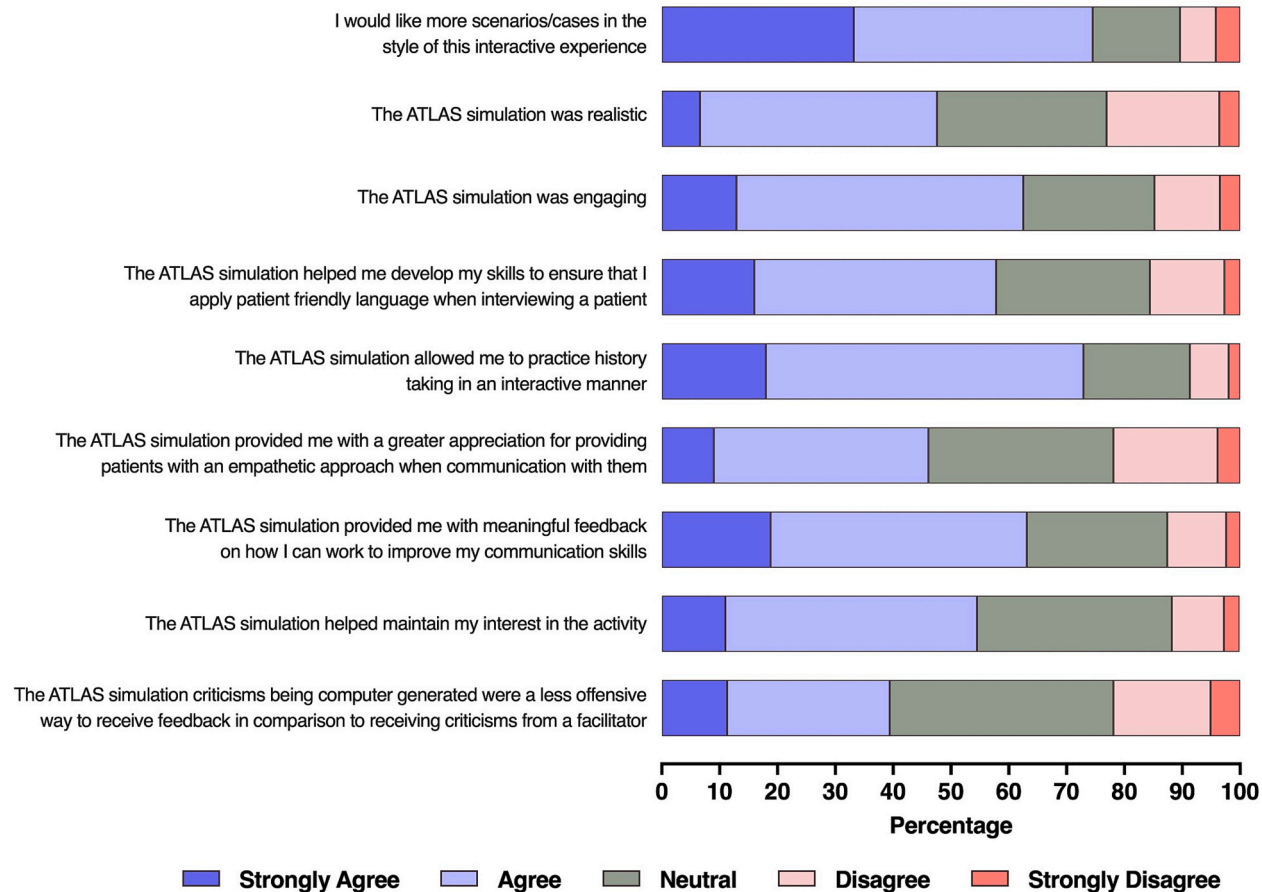


Fig. 6. Student perceptions of the ATLAS simulation experience. Stacked bar chart showing the distribution of agreement levels for nine statements about the ATLAS simulation ($n = 200$ ATLAS users who completed the survey). Responses range from “Strongly Agree” (dark blue) to “Strongly Disagree”. Percentages for each response category are shown on the x-axis. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
Illustrative quotes mapped to Fredricks' three dimensional Engagement Model.

Codes	Illustrative quotes from participants
Behavioural engagement	
Students wanted to utilise all possible methods to practice for the OSCE	"I appreciated that there were other methods to practice the OSCE" <i>Student 217</i>
Students built up their confidence	"ATLAS was a great tool for extra OSCE practice." <i>Student 234</i> "I felt more comfortable going in to the actual OSCE" <i>Student 211</i> "I practiced a LOT with friends, but I also used ATLAS. What didn't work well was reusing ATLAS scenarios because we knew what medications a patient was on subsequent attempts" <i>Student 212</i> "every aspect of practice was effective especially doing the cases given to us by the university and, ATLAS and workshop scenarios with friends" <i>Student 70</i>
To maximise OSCE preparation, you need to use a combination of peer-to-peer practice and ATLAS use	"A little uncomfortable but it still did the job" <i>Student 204</i>
Even though speaking to a robot was uncomfortable, students didn't want to miss out on practice that could lead to academic success	"I felt more comfortable practising on my own and in my own time student" <i>Student 207</i> "because I could do it in the comfort of my own home without the pressure of classmates watching" <i>Student 214</i>
Students who stated they were less extroverted appreciated the online own time privacy of ATLAS preparation	
Emotional engagement	
ATLAS was able to bring real-life interactions to the screen	"It felt like the real thing" <i>Student 8</i> "The interaction felt realistic" <i>Student 116</i> "Pretty amazing, as it can react to any of my questions" <i>Student 103</i> "I felt like it was a real scenario. However the long pauses before the response were a bit off-putting" <i>Student 239</i>
Technical hiccups can deter students from finishing off their attempts	"It was annoying how the patients speak at quite a slow pace. I also did not like how I had to keep scrolling up and down to get to fill out the medication history and then to press the record or end message button." <i>Student 213</i> "At first, I found it difficult to figure out what questions to ask an avatar" <i>Student 202</i>
Repeated use of ATLAS, improved emotional connection with the avatar	"At first, it was strange that the patient's facial expressions changed depending on what question I asked, but once I got use to this, I felt this aspect was good to have" <i>Student 236</i> "no computer can replicate a real patient" <i>Student 230</i> "It exceeded my expectations, though there are still many improvements needed for the AI software to be a better representation of the real interaction." <i>Student 99</i> "Sometimes the avatar would repeat the same thing and I wasn't sure if it was something important I needed to note" <i>Student 55</i> "You can't interrupt the avatar when they give you an unnecessarily long response, and hard to address each bit after" <i>Student 68</i> "My friends and I came up with our own patient scenarios with different personalities for each so it was as unique as possible" <i>Student 186</i>
The nuances of a real human interaction can be hard to replicate through AI	
ATLAS avatars often provided lengthy responses which overwhelmed students	
Cases are fixed, can make a better range with friends	
Cognitive engagement	
The personalised feedback allowed students to continually improve	"Motivated me to improve in more areas." <i>Student 238</i> "The feedback was good, it allowed me to pinpoint my weak points and reminded me of the questions I forgot to ask." <i>Student 200</i> "I think this was extremely beneficial as the more I practiced the more comfortable I got and it helped me find ways to improve my oral communication." <i>Student 209</i> "I feel like I am better adapting to the patient's responses, as I am better able to avoid "Ums" and "ahs" in my conversations. I remember previous oral assessments were plagued with these pondering statements. This in turn has shown I have improved in both my professionalism and oral communication skills." <i>Student 198</i> "Engaging with the interactive simulation's scenarios provoked a range of emotions, which ultimately gave me the chance to hone my professional skills." <i>Student 231</i> "Didn't tried it because it doesn't feel realistic to talk to AI instead of practising with your friends" <i>Student 254</i> "I don't really feel comfortable using AI in general so I avoid using it." <i>Student 223</i>
Communication and Professionalism was seen as the two skills ATLAS supported the most	
Students reported they were still not really comfortable or aware of AI tools and how they can benefit which limited their use of AI	

186.

Cognitive engagement

Despite some drawbacks, students generally found the feedback and practice scenarios useful for not only OSCE preparation but also for real life practice. Professional and communication skills were noted as the two skill areas where ATLAS provided the most support in improvement; 'I think this was extremely beneficial as the more I practiced the more comfortable I got and it helped me find ways to improve my oral communication.' *Student 209*; 'Engaging with the interactive simulation's scenarios provoked a range of emotions, which ultimately gave me the chance to hone my professional skills' *Student 231*. The personalised and honest feedback allowed for students to identify areas for improvement, e.g. clinical history taking; 'The feedback was good, it allowed me to pinpoint my weak points and reminded me of the questions I forgot to ask.' *Student 200*. Overall, the effort to engage with ATLAS seemed worthwhile, as it was seen to lead to mastery of skills needed in OSCEs and professional practice, particularly supported by the feedback loop ATLAS provides. Utilising AI tools for learning is still seen as a novel concept for this cohort of learners and some felt uncomfortable or unaware on how to use the tool; "Didn't tried it because it doesn't feel realistic to talk to AI instead of practising with your friends" *Student 254*; "I don't really feel comfortable using AI in general so I avoid using it." *Student 223*.

Feedback, identity and skill development

Students consistently described ATLAS as a platform that provided detailed, comprehensive feedback on their clinical conversations. One student reflected, *‘having an analysis of everything I was doing was just a really good experience.’* Student 194 This extensive feedback was perceived as extremely comprehensive, with him noting that it would be difficult for a human to capture all facets of communication within feedback, including elements such as appropriateness of hand gestures, eye contact, phrasing and the depth of clinical information delivered, in such detail. This enabled him to identify multiple areas to strengthen his clinical communication skills.

The ATLAS design was also described as human-like, with students emphasising the authenticity of the interaction and the safety of the environment. As one explained, *“the human-like experience, the judgement free feedback that we get”* Student 5 was central to his engagement. In this private, non-judgemental environment, he experimented with refining his interactions, tested phrasing and observed how the avatars responded. Each attempt built his confidence, reinforced his developing professional identity, and sharpened his communication skills. ATLAS was deemed to be a safe learning environment where students learned to embody the role of the pharmacist with developing confidence.

Discussion

Higher OSCE communication scores were associated with engaging in three or more ATLAS conversations relative to non-users (0 attempts). The observed improvements of 6.4–8.3 points among students completing 3+ conversations on the OSCE communication scale represent clinically meaningful changes that could translate to improved patient care skills. These improvements correspond to rank-biserial correlation of $r = 0.29$ – 0.41 (small to medium effects), which are particularly notable given already high baseline communication scores (mean = 86), suggesting that ATLAS was associated with refinement of communication subskills. Given each additional point on the rubric (Appendix 2) reflects the use of more verbal strategies to enhance interaction between interlocutors, small effect changes could lead to improved clinically significant changes within each subset of communication skills.

Notably, these significant improvements were achieved despite technical challenges with the pilot version of ATLAS, including system lag and overly formal avatar responses that deterred some students from completing their intended practice sessions. Future iterations addressing these technical limitations may yield even greater educational benefits. The use of AI tools in learning remains relatively new for this cohort and some students expressed discomfort or unfamiliarity with how to effectively use the tool. This hesitation could be related to what is commonly referred to as ‘AI hesitancy’, where individuals are uncertain or reluctant to engage with AI.^{23–25} Although public acceptance of GenAI has been mostly positive, concerns regarding academic integrity and plagiarism persist in educational settings. Initially, some institutions reacted by banning the technology outright. However, many have since shifted towards embracing AI in teaching and learning, implementing safeguards to ensure responsible use.^{6,7,9,26,27} Despite this, some students may remain hesitant due to concerns about the accuracy of its outputs, which cannot always be guaranteed or easily verified.^{23–25} Furthermore, the lack of institutional policies on the use of GenAI contributes to their uncertainty about whether AI use constitutes plagiarism.^{23–25} AI also raises concerns about privacy, with students wary that their personal information may be collected. Additionally, there are concerns about potential over-reliance on AI, which could hinder personal growth, skill development, and intellectual progress over time.

There are ongoing efforts to enhance AI literacy and introduce more AI-driven teaching and learning innovations within health profession curriculums. Against the wider evidence base, our findings align with recent reviews that describe promising but limited empirical support for AI/LLM enabled tools in health professions education.^{9,26–28} A recent systematic review²⁸ found that across AI interventions, most studies were single center with heterogeneous designs and sparse control of confounding; rigorous evidence for measurable educational outcomes remains weak, and standardized reporting frameworks are needed. Likewise, a rapid review of generative AI in undergraduate medical education²⁹ concluded that while use cases (virtual patients, decision making tutors) are proliferating, quantitative data demonstrating improved learner outcomes are still relatively scarce. Rahman and colleagues²⁴ highlight that trust, influenced by knowledge and prior experiences, plays a critical role in the adoption of emerging technologies. The advent of GenAI has transformed the feedback process by delivering personalised, real-time responses at scale. This technology reduces the burden on educators by automating the feedback process, enabling them to provide timely, individualised responses without extensive manual effort.²⁵ Students receive immediate, context-specific, and actionable guidance, leading to more effective and personalised learning outcomes. Feedback has long received much attention for its transformative potential in the learning process.³⁰ As an effective pedagogy, it promotes learning by creating cognitive dissonance and reinforcing knowledge, making it a vital component of clinical training programs.³¹ Feedback helps students recognise their strengths and identify areas for improvement to enhance future performance. GenAI has also been shown to offer an efficient and satisfactory approach to providing feedback to pharmacy students for their research ideas.³² Students highlighted that they valued the feedback provided by ATLAS, recognising its role in equipping them to step into the persona as a pharmacist and practice authentic professional interactions. What distinguished the platform was its attention to subtle communicative elements, such as hand gestures, eye contact and phrasing, which students experienced not simply as detail, but cues that made the feedback and interaction experience feel authentic. This realism likely fostered trust not only in the technology, but in their own developing ability to communicate effectively, encouraging them to experiment with different counselling approaches and refine their style with greater confidence. In this way, ATLAS served as a transformative learning experience, extending opportunities for feedback beyond the constraints of limited workshops, where students may be hesitant to practise in front of peers, and instead provide additional individualised, focused development opportunities for the development of clinical communication skills. Notably, female students benefited more from ATLAS than male students. These students are often identified as less extroverted and may value the privacy of practising online at their own pace, allowing them to improve their skills

before practising in front of others. Introverted students may find traditional face-to-face classrooms overwhelming and chaotic.³³ ATLAS offers these students an opportunity to engage with learning objectives more thoughtfully, free from classroom disruptions and the reluctance to ask questions or participate in class.³⁴ Interestingly, domestic students demonstrated consistently higher communication scores than international students across all OSCE components, though no significant interaction with ATLAS use was observed. This suggests that while international students from culturally and linguistically diverse backgrounds may face challenges such as feelings of isolation or exclusion from dominant cultural groups,³⁵ ATLAS did not differentially benefit this group compared to domestic students. Future iterations of ATLAS may need to incorporate culturally responsive design elements or language support features to better serve international students' unique learning needs. Using GenAI for professional communication in intercultural contexts presents challenges due to natural language processing limitations. Experts suggest a hybrid approach: combining human educators for personalised guidance and cultural understanding with AI-generated content to support the development of professional knowledge.³⁶

Despite the benefits of AI tools like ATLAS, 73.5% of students (based on survey data, $n = 200$) felt that in-person practice with peers or facilitators was more effective for developing empathy and nuanced communication skills. This observation is consistent with findings from the study investigating the use of OSCE Virtual Experience tool where students still preferred face-to-face practice for OSCEs.³⁷ Given that the OSCEs are conducted face-to-face, it is understandable that students view in-person practice as more authentic and beneficial in preparing for real-world interactions. It is crucial to emphasise that ATLAS was not designed for telehealth training, as the OSCE does not involve telehealth scenarios. Rather, ATLAS serves as a starting point for students to practice independently, helping them build confidence before transitioning to face-to-face sessions with peers or facilitators. In this regard, ATLAS proved to be a valuable supplementary practice tool.

An important consideration in interpreting these results is that the ATLAS user group demonstrated much higher baseline communication scores in their previous OSCE year (before ATLAS was available) compared to the control group (mean difference = 3.2 points, $p = 0.0011$). This pattern suggests that students who used ATLAS may have been inherently more motivated or academically stronger performers, representing a self-selection bias. To address this potential confounding factor, we employed beta regression analysis controlling baseline previous year's OSCE performance and key demographic variables (gender, domestic/international status). This statistical adjustment allowed us to isolate the specific impact of the ATLAS intervention from pre-existing student characteristics. Importantly, after controlling for these confounding factors, ATLAS usage remained significantly associated with improved communication scores in the primary care station ($\beta = 0.37$, 95% CI [0.13, 0.61], $p = 0.003$). The lack of significant effect in the history-taking station ($\beta = -1.10$, $p = 0.334$) may reflect the different skills required for longer, more complex medication reconciliation interviews compared to focused counselling scenarios. The persistence of significant ATLAS effects in primary care after controlling for baseline performance and demographics provides evidence that the intervention delivers genuine educational value beyond what can be attributed to user self-selection. However, we acknowledge important limitations in causal inference. While our beta regression models controlled for measured confounders, unmeasured variables such as intrinsic motivation, study habits, or learning preferences may still differ between ATLAS users and non-users.

The quasi-experimental nature of this study, where ATLAS use was optional rather than randomly assigned, limits our ability to make definitive causal claims. Future research employing randomised controlled trial designs, where students are randomly assigned to ATLAS access or control conditions, would help isolate the causal impact of ATLAS from confounding student characteristics and strengthen inference about the tool's effectiveness.

Although our results suggest ATLAS use is associated with improvement in OSCE communication grades, we do not know whether students used any other form of support prior to the assessment that could have impacted their performance. We did not collect data on additional support seeking behaviors (e.g., private tutoring, external study groups), which represents a limitation of this study. However, given both cohorts had access to the same standard preparation resources and peer practice opportunities, it is reasonable to assume that any additional support methods were likely balanced between the control and the ATLAS groups. If ATLAS users were more likely to seek additional support, our estimates may overstate the specific effect of ATLAS; conversely, if non-users sought more alternative preparation methods to compensate for not using ATLAS, our estimates may be conservative. The direction and magnitude of this potential bias cannot be determined from our data. ATLAS is also best used outside of the classroom to utilise its camera functions without outside distraction so we do not know if students used ATLAS in different ways. We do have students' conversational transcripts with ATLAS and plan to investigate use in future research. Future research could also evaluate the cost of ATLAS versus its face-to-face substitute. Furthermore, initial student feedback about technical issues such as system lag and overly formal responses has already been incorporated into newer iterations of the platform, creating more natural interactions. Whilst there has been criticism that open GenAI has led to poor learning outcomes, this study shows the successful use of an educator-guided GenAI tool for clinical communication and also shows links to improvement in academic grades, particularly benefitting those who are hesitant to practise role-playing in the classroom. With the same model template harnessing repeated practice and educator guided feedback, ATLAS could be adopted to improve communication skills across other healthcare disciplines to prepare students for workforce conversations.

Ethics approval

This project was approved by the institution's Human Research and Ethics Committee (Project no: 41702).

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Declaration of generative AI and AI-assisted technologies:

GenAI was used to assist with data analysis for thematic coding. The first round of coding was conducted using Python scripts with OpenAI API integration to elicit the most common themes. Two researchers (AM, AL) reviewed the coding conducted by the AI together and mapped the themes manually.

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Declaration of competing interest

The ATLAS platform was developed at Monash University. While future commercialisation could constitute a competing interest for JM and AM, it is currently in non-commercial use and the authors derive no financial benefit.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cptl.2026.102675>.

Data availability statement

The data that support the findings of this study are available from the corresponding author, [AL], upon reasonable request.

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